Infectious vs. Non-Communicable Disease Burden Across Regions and Income Levels: A Global Analysis (2000–2021)

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https://manmanhe1014.github.io/JSC370-Infectious-vs.-Non-Communicable-Disease-Burden-Across-Regions-and-Income-Levels/

1 Introduction

The global burden of disease has shifted significantly over the past century. As countries develop economically, they tend to undergo the epidemiological transition. Epidemiological transition is a shift from predominantly infectious diseases (such as tuberculosis and malaria) toward non-communicable diseases (NCDs) like cardiovascular disease, diabetes, and cancer. However, this transition is neither universal nor linear. Many low- and middle-income countries now face a "double burden," contending with high rates of both infectious and non-communicable diseases simultaneously.

This project investigates how disease burden patterns vary across 178 countries from 2000 to 2021, and how these patterns relate to economic development. We address two research questions:

1. How do infectious versus non-communicable disease patterns differ across WHO regions and World Bank income groups?
2. Which country-level factors — GDP per capita, health expenditure, urbanization — best explain that variation?

We combine health indicators from the WHO Global Health Observatory with economic data from the World Bank, and apply XGBoost models for both classification (predicting a country's income group from its disease profile) and regression (predicting life expectancy from disease and economic indicators). The analysis reveals that disease burden is an important signal of economic status, and that the relationship between health spending and outcomes is more nuanced than spending levels would suggest.

2 Methods

2.1 Data Acquisition and Cleaning

Health indicators were obtained from the WHO Global Health Observatory API (https://ghoapi.azureedge.net/api/), including life expectancy at birth, TB incidence per 100,000, malaria deaths, new HIV infections per 1,000, and NCD mortality as a percentage of total deaths. Economic indicators were retrieved from the World Bank API (https://api.worldbank.org/v2/), including GDP per capita (PPP), health expenditure as a percentage of GDP, urban population percentage, and total population.

The two data sources were merged by ISO country code and year, covering 178 countries over 22 years (2000–2021). Records with missing values in key indicators were removed. The final cleaned dataset contains observations across all World Bank income groups (Low income, Lower middle income, Upper middle income, and High income) and six WHO regions.

2.2 Feature Engineering

Several derived features were created to capture disease burden more effectively:

- **HIV per 1,000**: normalized HIV infections by population.
- **Malaria per 100,000**: normalized malaria deaths by population.
- **Log GDP**: log-transformed GDP per capita to reduce right skew.
- **TB-to-NCD ratio**: ratio of TB incidence to NCD mortality percentage, capturing the relative infectious vs. non-communicable burden.
- **Infectious burden score**: composite index combining TB incidence, HIV rate ($\times 100$), and malaria death rate.

2.3 Exploratory Data Analysis

Interactive visualizations were developed to explore spatial and temporal patterns (available on the project website). These include a choropleth map of life expectancy over time, a sunburst chart showing the hierarchical structure of TB burden by WHO region, income group, and country, and a Gapminder-style animated bubble chart tracking GDP, life expectancy, and population from 2000 to 2021.

2.4 Modeling

All modeling was performed on 2019 cross-sectional data (pre-COVID) using 10 features. The dataset was split 70/30 into training and test sets with stratification (for classification).

2.4.1 XGBoost Classification

We trained an XGBoost classifier to predict World Bank income group (4 classes) from disease and economic indicators. XGBoost minimizes a regularized objective:

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \left[\gamma T_k + \frac{1}{2}\lambda \|w_k\|^2 + \alpha \|w_k\|_1 \right] \tag{1}$$

where ℓ is the multi-class log-loss, T_k is the number of leaves in tree k, and λ, α, γ are regularization hyperparameters.

A baseline model was trained with default hyperparameters, then refined through randomized search over 30 candidate configurations using 5-fold stratified cross-validation. The search explored tree complexity, learning rate, and regularization strength to balance model flexibility against overfitting. Early stopping on a held-out validation set determined the final number of boosting rounds.

Evaluation metrics include accuracy, macro F1-score, per-class precision/recall, confusion matrix, and one-vs-rest ROC/AUC curves.

2.4.2 XGBoost Regression

We trained an XGBoost regressor to predict life expectancy at birth using the same feature set and tuning procedure as above, optimizing for root mean squared error. Early stopping was again used to prevent overfitting.

Evaluation metrics include RMSE, R^2, and MAE. Diagnostic plots (predicted vs. actual and residual distribution) were used to assess model fit.

3 Results

3.1 Exploratory Findings

The interactive visualizations (see project website) reveal that life expectancy gains from 2000 to 2021 were widespread but uneven, with sub-Saharan Africa showing the most dramatic improvements. The sunburst chart confirms that Africa dominates global TB burden, and the animated bubble chart reveals a visible COVID-19 dip in 2020.

Three interactive figures are available on the project website:

- **Figure 1 (Choropleth Map):** Life expectancy at birth by country, animated across 2000–2021.
- **Figure 2 (Sunburst Chart):** Hierarchical TB burden by WHO region, income group, and country (2019).
- **Figure 3 (Bubble Chart):** GDP vs. life expectancy animated over time, sized by population.

3.2 Classification Results

Table 1 summarizes the classification performance. The tuned model achieved 88.9% accuracy and 0.887 macro F1 on the test set, with a macro AUC of 0.973.

Table 1: XGBoost classification performance: baseline vs. tuned model.

Metric	Baseline	Tuned
Accuracy	87.0%	88.9%
Macro F1	0.871	0.887
Macro AUC	—	0.973

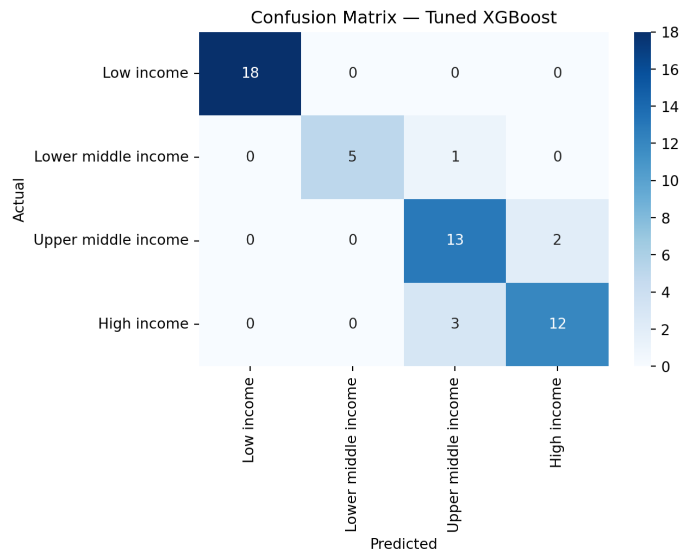

Figure 1: Confusion matrix for the tuned XGBoost classifier. Low-income countries are classified with perfect accuracy, while the most confusion occurs between upper middle and high income groups, reflecting their overlapping disease and economic profiles.

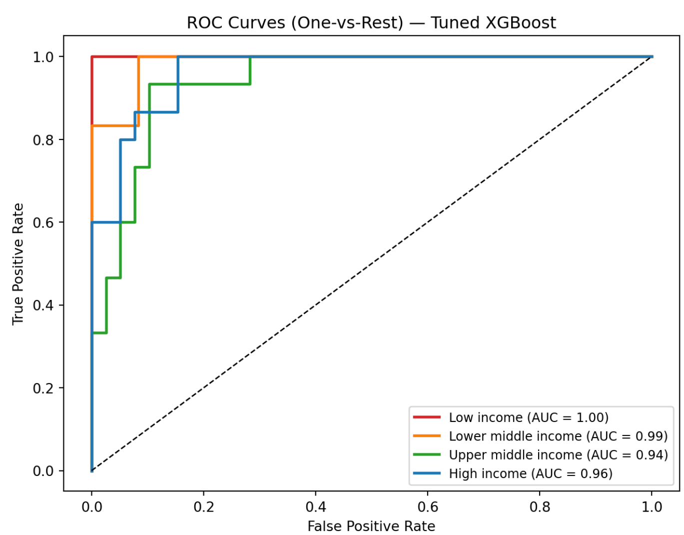

Figure 2: One-vs-rest ROC curves for the four income groups. Low income achieves a perfect AUC of 1.00, followed by lower middle income (0.99), high income (0.96), and upper middle income (0.94), indicating that countries at the extremes of the income spectrum are easiest to distinguish.

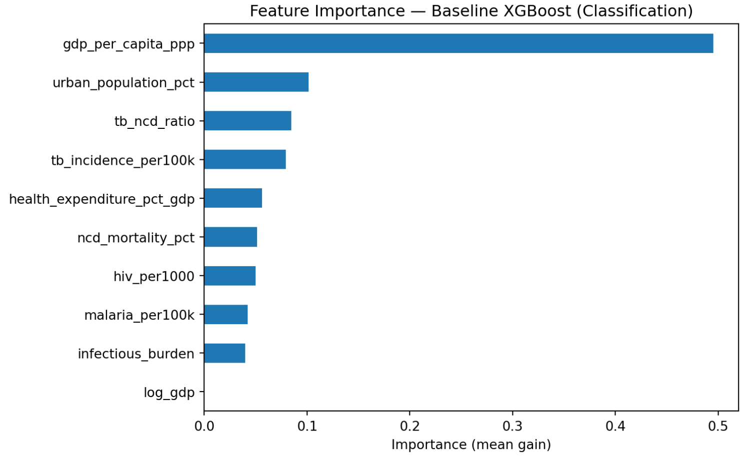

Figure 3: Feature importance (mean gain) from the XGBoost classifier.

3.3 Regression Results

Table 2 summarizes regression performance. The tuned model predicts life expectancy with an RMSE of 1.78 years and R^2 of 0.9230.

Table 2: XGBoost regression performance: baseline vs. tuned model.

Metric	**Baseline**	**Tuned**
RMSE (years)	2.00	1.78
R^2	0.9032	0.9230
MAE (years)	—	1.21

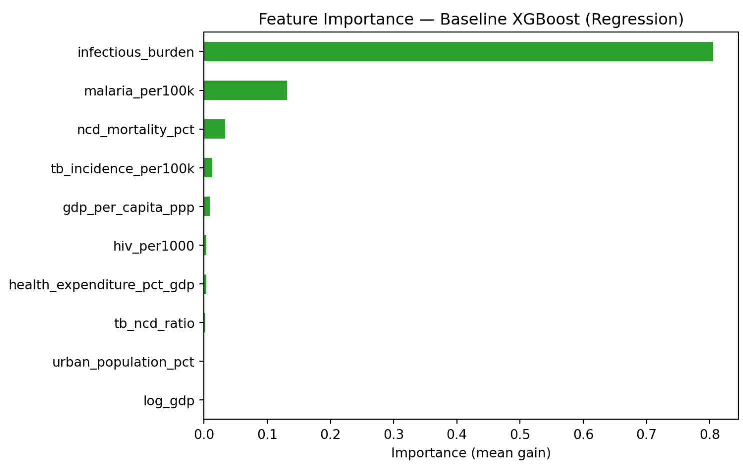

Figure 4: Feature importance (mean gain) from the XGBoost regressor. Infectious burden score and malaria death rate are the strongest predictors of life expectancy.

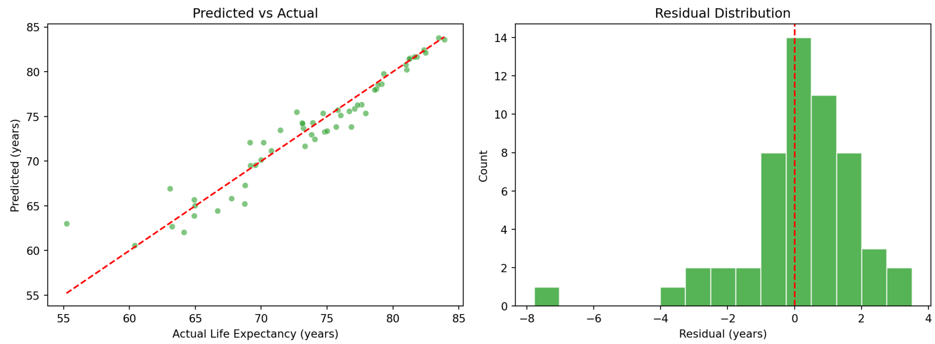

Figure 5: Left: predicted vs. actual life expectancy. Right: residual distribution centered near zero.

3.4 Partial Dependence

Partial dependence plots (Figures 6 and 7) show the marginal effect of top features on predictions.

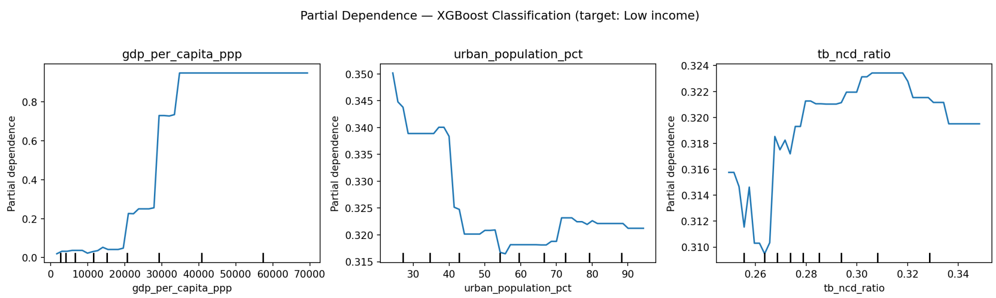

Figure 6: Partial dependence plots for top 3 classification features (target: Low income).

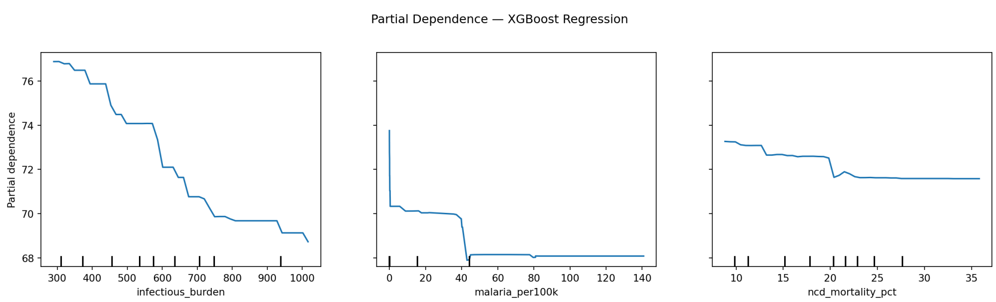

Figure 7: Partial dependence plots for top 3 regression features.

4 Conclusions and Summary

This analysis reveals that low-income countries face a genuine "double burden" of both infectious and non-communicable diseases. This challenges the classical epidemiological transition model. Disease burden proves to be a powerful economic signal, as the classifier distinguishes income groups with 88.9% accuracy and 0.973 AUC. Life expectancy is highly predictable ($R^2 = 0.923$), with the infectious burden composite and malaria death rate emerging as the strongest predictors, surpassing GDP and health spending. Moreover, health expenditure as a share of GDP shows weak association with outcomes, which suggests resource allocation matters more than total spending. All income groups have seen improvements from 2000 to 2021, but the gap between low- and high-income countries persists, and COVID-19 caused a visible setback in 2020.

4.1 Limitations

There are several limitations. First, the modeling is cross-sectional (2019 only), so causal claims cannot be made. Second, data quality varies across countries, particularly for low-income nations with weaker health surveillance systems. Third, the feature set does not capture important factors such as governance, conflict, or healthcare infrastructure quality. Finally, for the XGBoost models, are less interpretable than simpler models. Although it is good for prediction, the partial dependence plots provide some insight but do not capture interaction effects fully.